Calculus II

Solution to Assignment 7

Part I. Regular Problems

1. (a) $0 \le r < 1$, any θ.
   (b) $0 < r$ and $0 < \theta < \frac{\pi}{2}$.

2. (a) (iii) (b) (i) (c) (ii) (d) (iv)

3. $f(x,y,z) = \sqrt{x^2+y^2+z^2} \Rightarrow f_x(x,y,z) = \frac{x}{\sqrt{x^2+y^2+z^2}}, f_y(x,y,z) = \frac{y}{\sqrt{x^2+y^2+z^2}}$, and $f_z(x,y,z) = \frac{z}{\sqrt{x^2+y^2+z^2}}$, so $f_x(3,2,6) = \frac{3}{7}, f_y(3,2,6) = \frac{2}{7}, f_z(3,2,6) = \frac{6}{7}$. Then the linear approximation of f at $(3,2,6)$ is given by

$$\begin{aligned} f(x,y,z) &\approx f(3,2,6) + f_x(3,2,6)(x-3) + f_y(3,2,6)(y-2) + f_z(3,2,6)(z-6) \\ &= 7 + \frac{3}{7}(x-3) + \frac{2}{7}(y-2) + \frac{6}{7}(z-6) = \frac{3}{7}x + \frac{2}{7}y + \frac{6}{7}z \end{aligned}$$

   Thus, $\sqrt{(3.02)^2 + (1.97)^2 + (5.99)^2} = f(3.02, 1.97, 5.99) \approx \frac{3}{7}(3.02) + \frac{2}{7}(1.97) + \frac{6}{7}(5.99) \approx 6.9914$.

4. (a) $y; x; 1; 1$
   (b) $\frac{\partial f}{\partial x} = \frac{1}{y}$ and $\frac{\partial f}{\partial y} = -\frac{x}{y^2}$; Evaluated at $(1,1)$:

$$\frac{\partial f}{\partial x} = \boxed{1} \quad \text{and} \quad \frac{\partial f}{\partial y} = \boxed{-1}$$

   (c) $6x; 4y; 6; 8$
   (d) $\frac{\partial z}{\partial x} = \cos(x^2 - 3xy)(2x - 3y)$ and $\frac{\partial z}{\partial y} = \cos(x^2 - 3xy)(-3x)$; Evaluated at $(\sqrt{\pi}, \sqrt{\pi})$:

$$\frac{\partial z}{\partial x} = \boxed{-\sqrt{\pi}} \quad \text{and} \quad \frac{\partial z}{\partial y} = \boxed{-3\sqrt{\pi}}$$

5. (a) $yz; xz; xy; 1; 1; 1$

   (b) $f_x = \frac{x}{\sqrt{x^2+y^2+z^2}}, \quad f_y = \frac{y}{\sqrt{x^2+y^2+z^2}}, \quad f_z = \frac{z}{\sqrt{x^2+y^2+z^2}}$; Evaluated at $(3, 0, 4)$:

$$f_x = \boxed{\frac{3}{5}}, \quad f_y = \boxed{0}, \quad f_z = \boxed{\frac{4}{5}}$$

   (c) $-y^2 \sin\left(xy^2\right) + 3yze^{3xyz}; -2xy \sin\left(xy^2\right) + 3xze^{3xyz}; 3xye^{3xyz}; 3e^{3\pi}; \ 3\pi e^{3\pi}; 3\pi e^{3\pi}$

   (d) $f_x = yzx^{yz-1}, \quad f_y = x^{yz} \ln(x) \cdot z, \quad f_z = x^{yz} \ln(x) \cdot y$; Evaluated at $(1, 1, 0)$:

$$f_x = \boxed{0}, \quad f_y = \boxed{0}, \quad f_z = \boxed{0}$$

Part II. Group Problems

1. (a) $f_x(x, y) = -2x\mathrm{e}^{-x^2-y^2}, f_y(2, 0) = 0,$

   (b) $x\mathrm{e}^y + \mathrm{e}^x$

   (c) $-y \sin(xy) + \cos(xy) - xy \sin(xy)$

2. Note that $u = x - a, v = y - b$ and

$$(a + u)^2 = a^2 + 2au + u^2, \quad (b + v)^3 = b^3 + 3b^2v + 3bv^2 + v^3.$$

   Therefore,

$$\begin{aligned} f(x, y) = f(a + u, b + v) &= (a^2 + 2au + u^2)(b^3 + 3b^2v + 3bv^2 + v^3) \\ &= a^2b^3 + 2ab^3u + 3a^2b^2v + \cdots \\ &= a^2b^3 + 2ab^3(x - a) + 3a^2b^2(y - b) + \cdots \end{aligned}$$

   Identify c_1, c_2, c_3:

$$c_1 = a^2b^3, \ c_2 = 2ab^3, \ c_3 = 3a^2b^2.$$

   (a) Expression for c_1:

$$c_1 = \boxed{f(a, b)}$$

   (b) Expressions for c_2 and c_3:

$$c_2 = \boxed{\frac{\partial f}{\partial x}(a, b)}, \quad c_3 = \boxed{\frac{\partial f}{\partial y}(a, b)}$$

(c) Functions $\ell(u,v)$ and $s(u,v)$:

$$\ell(u,v) = 2ab^3u + 3a^2b^2v,$$

and

$$\begin{aligned} s(u,v) &= f(a+u,b+v) - f(a,b) - \ell(u,v) \\ &= (a^2+2au+u^2)(b^3+3b^2v+3bv^2+v^3) - (a^2b^3+2ab^3u+3a^2b^2v) \\ &= a^2v^2(3b+v) + 2auv(3b^2+3bv+v^2) + u^2(b^3+3b^2v+3bv^2+v^3). \end{aligned}$$

We next show that

$$\frac{|s(u,v)|}{\|(u,v)\|} \to 0, \quad \text{as } (u,v) \to (0,0).$$

Since we are concerned with $(u,v) \to (0,0)$, we only need to consider very small u, v, say

$$|u| \le 1, \ |v| \le 1.$$

In this case, we have

$$\begin{aligned} |s(u,v)| &\le a^2v^2(3|b|+1) + 2|a||uv|(3b^2+3|b|+1) + u^2(|b|^3+3b^2+3|b|+1) \\ &\le \kappa(v^2+2|uv|+u^2) \\ &\le 2\kappa(v^2+u^2), \end{aligned}$$

where

$$\kappa := \max\{a^2(3|b|+1), |a|(3b^2+3|b|+1), |b|^3+3b^2+3|b|+1\} > 0$$

is a constant. Therefore, for any $\epsilon > 0$, we can choose $\delta = \epsilon/(2\kappa)$. Then if $\|(u,v)\| < \delta$, we get

$$\frac{|s(u,v)|}{\|(u,v)\|} \le 2\kappa \frac{v^2+u^2}{\sqrt{v^2+u^2}} = 2\kappa\|(u,v)\| < \epsilon,$$

showing that

$$\frac{|s(u,v)|}{\|(u,v)\|} \to 0, \quad \text{as } (u,v) \to (0,0).$$